INCH-POUND

A-A-56023C
21 July 2020
SUPERSEDING
A-A-56023B
5 November 2010

COMMERCIAL ITEM DESCRIPTION

INSULATING COMPOUND, ELECTRICAL, SILICONE RUBBER, ROOM TEMPERATURE VULCANIZING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers four types of room temperature vulcanizing, silicone rubber electrical insulating compounds. The insulating compounds are intended for use in potting and encapsulating electrical and electronic components.

2. CLASSIFICATION. The insulating compounds shall be classified by the following types, as specified (see (7.4(b)):

Type I	- Viscosity, 200 to 500 poises
Type II	- Viscosity, 200 to 600 poises
Type III	- Viscosity, 300 to 900 poises
Type IV	- Viscosity, 80 to 250 poises

3. SALIENT CHARACTERISTICS

3.1 General Requirements. The insulating compounds shall consist of an organo-polysiloxane polymer which, when mixed with a catalyst, shall cure to a rubbery solid. Cure shall be affected by catalytic action, not solvent evaporation. The insulating compounds shall have a shelf life of not less than six months when stored in the original unopened containers at a temperature not greater than 26.7 ºC. The glass transition temperature (tg), or softening point, shall be as specified by the manufacturer when tested in accordance with IPC-TM-650, method 2.4.24, "IPC Test Methods Manual, Glass Transition Temperature and Z-Axis Thermal Expansion by TMA".

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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The insulating compound materials shall be homogeneous and free of foreign materials, and shall be packaged in clean well-sealed containers, in capacities specified by the contract. The physical requirements of the uncured insulating compounds are shown in table I. The physical and electrical requirements of the cured insulating compounds are shown in table II. Conditioning requirements and testing conditions are shown in table III. Test methods are shown in table IV.

TABLE I. Physical requirements, uncured insulating compounds.

Requirement	Conditioning procedure[1] (see table III)	Requirement values			
		Type I	Type II	Type III	Type IV
Uncatalyzed, viscosity, average, minimum/maximum, poises	A	200/500	200/600	300/900	80/250
Catalyzed, pot life, minimum, hours	A	0.5	0.5	0.5	0.5

[1] Conditioning procedure A: As received, no special conditioning. Conditioning procedure E: Temperature conditioned (see table II, footnotes 1, 2 and 3).

TABLE II. Physical and electrical requirements, cured insulating compounds.

Requirement	Conditioning procedures[1 2 3] (see table III)	Requirement values, all types
Hardness shore A, average minimum/maximum	A E-70/232 °C + E-1/-24 °C E-1/-54 °C + E-1/24 °C	35/70 35/70 35/70
Tensile strength, average, minimum, pounds force per square inch (lbf/in^2)	A E-70/232 °C + E-1/24 °C E-1/-54 °C + E-1/24 °C	300 150 300
Tear strength, average minimum, pounds force per inch of thickness (lbf/in)	A	15
Elongation, average minimum, percent	A E-70/232 °C + E-1/-54 °C	100 75 100
Dielectric strength, average, minimum, volts per mil, approx. 75 mil sheet	A	350

[1] Proceed with conditioning procedures after curing insulating compounds for not less than 72 hours at 24 °C ±3 °C.
[2] Conditioning procedure A: As received, no special conditioning.
[3] Conditioning procedure E: Temperature conditioned; encoding definition is a capital letter indicating type of conditioning, a number indicating the duration of the conditioning in hours, and a number indicating the temperature in degrees C. A sequence of conditions is denoted by a "+" sign between conditions, see ASTM D618, "Standard Practice for Conditioning Plastics for Testing".

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TABLE III. Conditioning requirements and testing conditions.

Conditioning procedure	Conditioning tolerances[1]		Testing conditions	
	Time	Temp	Temperature and tolerances	Remarks
E (low temperature)	−0, +6 min.	2.8 °C	24° ±5.6 °C or -54° ±2.8 °C as indicated	Start test immediately after the cycling has been completed.
E (high temperature)	−0, +6 min.	5.6 °C	24° ±5.6 °C or 232° ±5.6 °C as indicated	Start test immediately after the cycling has been completed.

[1] Condition A: As received, no special conditioning. Condition E: Temperature conditioned.

TABLE IV. Test methods.

Requirement	Test method
Viscosity	ASTM D1084, method B
Pot life	MIL-PRF-23586
Hardness	ASTM D2240
Tensile strength	ASTM D412
Tear strength	ASTM D624
Elongation	ASTM D412
Dielectric strength	ASTM D149
Glass transition temperature	IPC-TM-650, method 2.4.24

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

5.3 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, "Preferred Metric Units for General Use by the Government", and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the

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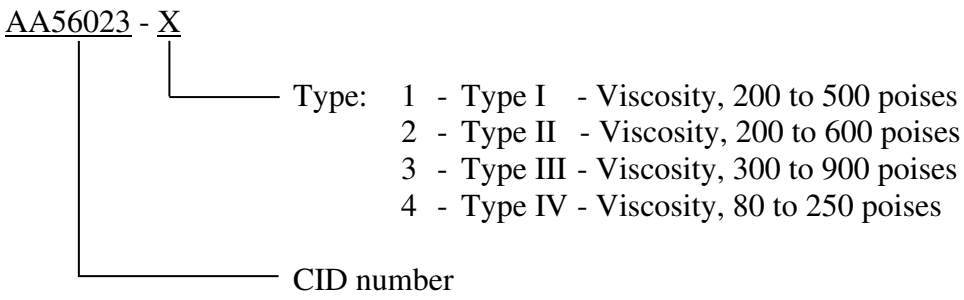
tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.4(c)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



Example of reference part number: AA56023-1

7.2 Safety data sheets (SDS). Safety data sheets shall be prepared by Manufacturers for the insulating compounds described herein in accordance with the requirements of FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication". Contracting officers will identify those activities requiring copies of completed safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

7.3 Sources of documents.

7.3.1 FAR. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/.

7.3.2 Federal standards. Electronic copies of federal standards may be obtained from https://assist.dla.mil/.

7.3.3 Military specifications. Electronic copies of military specifications may be obtained from https://assist.dla.mil/.

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7.3.4 ASTM standards. Electronic copies of ASTM standards may be obtained from https://www.astm.org/.

7.3.5 IPC standards. Electronic copies of IPC standards may be obtained from http://www.ipc.org/.

7.4 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type (see 2).
c. Preservation, packing, and marking (see 6.1).

7.5 Subject term (key word) listing.

Catalytic
Conditioning
Cure
Dielectric
Electronic
Encapsulating
Poise
Polymer
Polysiloxane
Potting
Viscosity

MILITARY INTERESTS:

Custodians:
Army - MI
DLA - GS

Review Activities:
Army - CR4, MD, MR

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS8

(Project 5970-2020-002)

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